

BOULDER 14 W, CO, USA

WMO: 745200

Lat: 40.0354N

Lon: 105.5409W

Elev: 2996

StdP: 70.15

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 94075

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-21.2	-18.4	-27.4	0.4	-13.8	-25.1	0.6	-12.2	11.2	-6.4	10.7	-5.5	1.9	N/A	0.687

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	21.9	9.0	20.6	8.9	19.5	8.7	11.9	16.2	11.2	16.2	10.5	16.1	2.1	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.5	11.5	13.3	9.5	10.7	12.7	8.4	10.0	12.1	43.9	16.3	41.5	16.3	39.3	15.9	15.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.9	8.8	7.7	-27.5	24.4	3.7	1.1	-30.2	25.2	-32.3	25.8	-34.4	26.4	-37.1	27.1		
(5)			-28.2	13.9	4.3	0.7	-31.3	14.4	-33.8	14.8	-36.3	15.2	-39.4	15.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.0	-6.4	-7.0	-3.1	-0.5	3.8	9.9	12.8	11.6	8.5	2.5	-2.3	-6.7		
	DBStd	8.34	5.23	5.59	5.16	4.61	4.18	3.27	2.36	2.39	3.57	4.94	5.49	5.74		
	HDD10.0	3143	507	475	406	314	195	42	6	13	67	232	369	518		
	HDD18.3	5972	766	708	664	564	452	254	173	209	296	489	619	776		
	CDD10.0	214	0	0	0	0	1	38	92	61	21	1	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	6	0	0	0	0	0	2	3	1	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	2.7	4.0	3.7	3.1	2.5	2.0	1.9	1.7	1.7	1.9	2.7	3.3	3.8		
(15) Precipitation	PrecAvg	470	30	28	38	51	51	39	51	50	44	30	30	28		
	PrecMax	734	78	62	114	119	126	83	115	97	314	55	50	63		
	PrecMin	332	10	7	3	5	18	8	1	19	0	7	13	6		
	PrecStd	86	15	14	22	27	25	21	27	20	56	14	9	13		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	7.0	11.8	13.6	18.3	23.4	23.7	21.9	21.2	16.4	11.0	6.5		
		MCWB	-1.2	-0.8	1.8	3.2	5.6	8.8	9.4	9.0	7.8	5.0	1.6	-1.8		
	2%	DB	4.4	4.8	9.2	11.5	16.1	21.4	22.2	20.6	19.0	14.3	8.8	4.1		
		MCWB	-2.2	-1.8	0.8	2.5	5.2	8.2	9.4	9.0	7.8	4.3	0.6	-2.8		
	5%	DB	2.6	2.6	6.9	9.5	14.1	19.9	21.0	19.5	17.6	12.6	7.1	2.4		
		MCWB	-3.1	-2.9	-0.5	1.3	4.4	7.8	9.6	8.8	6.8	3.7	-0.1	-3.3		
	10%	DB	1.0	0.7	4.9	7.3	12.0	18.1	19.7	18.2	16.1	10.5	5.3	1.0		
		MCWB	-3.8	-3.9	-1.4	0.4	3.9	7.4	9.5	8.7	6.6	2.9	-1.3	-3.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.2	0.1	2.5	4.3	7.6	12.3	13.3	12.4	11.5	7.0	2.4	-0.8		
		MCDB	5.4	5.5	10.6	11.4	13.1	16.4	17.2	16.2	16.2	12.4	9.6	4.3		
	2%	WB	-1.5	-1.2	1.2	3.0	6.1	10.5	12.3	11.4	9.9	5.6	1.1	-1.7		
		MCDB	2.7	4.3	8.6	10.1	11.8	16.2	16.6	16.2	14.7	11.2	7.6	2.6		
	5%	WB	-2.4	-2.4	0.0	1.9	5.1	9.7	11.7	10.6	8.7	4.5	0.1	-2.5		
		MCDB	2.0	2.3	6.3	8.0	11.6	16.5	16.3	16.3	14.4	10.7	6.0	1.7		
	10%	WB	-3.4	-3.2	-1.0	0.9	4.2	8.8	10.9	10.0	7.9	3.4	-0.9	-3.3		
		MCDB	0.7	0.9	4.4	6.2	11.0	16.0	16.1	15.9	14.0	9.6	4.5	0.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	8.4	9.4	9.4	10.6	13.4	12.6	12.6	12.4	10.1	8.6	8.0		
		MCDBR	9.8	9.2	11.4	12.9	15.0	15.9	14.8	14.3	14.6	13.4	10.2	8.8		
		MCWBR	6.0	5.3	5.6	6.3	6.9	6.9	6.4	6.4	6.4	6.3	5.6	5.2		
	5% WB	MCDBR	9.1	7.9	10.7	12.0	13.2	14.0	12.2	12.4	12.1	11.8	10.2	8.4		
		MCWBR	5.9	5.0	5.6	6.4	7.1	7.6	6.8	6.5	6.3	6.4	5.5	5.5		
(40) Clear-Sky Solar Irradiance	taub		0.206	0.212	0.230	0.256	0.270	0.276	0.298	0.295	0.259	0.238	0.219	0.210		
	taud		2.611	2.581	2.577	2.567	2.607	2.605	2.571	2.563	2.686	2.712	2.680	2.626		
	Ebn at Noon		1012	1043	1044	1024	1008	997	971	970	997	998	989	983		
	Edh at Noon		65	78	88	96	95	95	98	96	79	68	60	59		
(44) All-Sky Solar Radiation	RadAvg		2.63	3.67	4.86	5.52	5.96	6.94	6.29	5.59	5.03	3.73	2.67	2.22		
	RadStd		0.12	0.20	0.41	0.37	0.49	0.39	0.30	0.28	0.32	0.25	0.17	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page